

Linked List – Stack / Queue

Due Date : Monday October 16th , 2023

Sections 001 & 008 , No Later than 10:00 am

Write a C++ program that will simulate an ADT stack / queue **using a linked List ADT with actual pointers**. The program will perform the following operation in sequences :

- A. Build a stack that consist of 5 random capital letters that are $\geq$ 'C' and $\leq$ 'K'
- B. Build a queue that consist of 5 capital letters that are $\geq$ 'H' and $\leq$ 'S'
- C. Insert the element 'W' in the stack (LIFO).
- D. Insert the element 'X' in the stack (LIFO).
- E. Remove an element from the stack (LIFO).
- F. Insert the element 'X' in the queue (FIFO).
- G. Insert an element 'A' at in the 2nd location in the queue.
- H. Remove the first element from the queue (FIFO).
- I. Remove the 4th element from the queue.
- J. Find Min element in the queue, and Max element in the queue.
- K. Delete the entire stack.
- L. Remove 'W' from the stack . Error message should be displayed if the stack is empty.
- M. Delete the entire queue.
- N. Remove an 'X' from the queue . Error message should be displayed if the queue is empty.
- X. Exit the program.

The entire list must be displayed After completion of A – I , K, and M. The program must continue running until the letter X is entered.

Notes :

- Validation must be done with the repeat process. Only **capital** letters are accepted. Everything else should be rejected.
- Must use at least 14 functions to handle the operations (A – N)
- Must place the following header / signature line before terminating the program.
- You are not allowed to use any types of arrays, any type of global arrays. You are also not allowed to use global variables.

Linked List , Stack , Queue Operations

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**LL / Stack & Queue Program is Implemented By :
Husain Gholoom - October 16th , 2023**

Style Guidelines:

At the beginning of your program (and before the #include statement), include the following :

Header comments (file documentation block) should be at the top of each file and should contain Author / s, Due Date, Assignment Number, Course number and section, Instructor, and a brief description of the purpose of the code in the file. For example :

```
//  
//  Author : (Your name/s here!!)  
//  
//  Serial Number / s :  
//  
//  Programming Assignment Number 4  
//  
//  Fall 2023 - CS 3358 - Your Section Number  
//  
//  Due Date :  
//  
//  Instructor:  Husain Gholoom.  
//  
//  <Brief description of the purpose of the program>
```

Variable names :

- Must be meaningful.
- The initial letter should be lowercase, following words should be capitalized, no other caps or punctuation (i.e. `weightInPounds`).
- Each variable must be declared on a separate line with a descriptive comment.

Named constants :

- Use for most numeric literals.
- All capitals with underscores (i.e. TX_STATE_SALES_TAX)
- Should occur at top of function, or global (only if necessary)

Line length of source code should be no longer than 80 characters (no wrapping of lines).

Indentation :

- Use 2-4 spaces (but be consistent throughout your program).
- Indent blocks, within blocks, etc.
- Use blank lines to separate sections.

Comments for variables :

All variable definitions should be commented as follows:

```
int  gender;      // integer value for the gender,  
                  // 1 = Male , 2 = Female ,
```

Rules :

1. Your program **must compile**. The program will be tested using Code::Blocks version 20.03 for windows.
2. The entire program must be **documented according to the style above**.
3. You must use the appropriate libraries in writing this program, however, you are **not allowed** to use libraries that support the functionalities of linked lists such as list or forward_list. You are also not allowed to use libraries that support stacks and queues. Finally, are also not allowed to use recursion.
4. **Your program must use a class definition along with appropriate member functions when applicable (LL notes page 22) . You are going to need to use 2 classes, one for the stack and one for the queue.**

```
struct Node
{
    char data;
    Node *link;
};
```

5. You must name your program as :

○ **PG4_F2023_3358_1_LastName_FirstName.cpp (not .cbp)**

Where LastName is your Last Name and FirstName is your First Name.

6. No late assignments will be accepted. No extension will be provided.

Use Canvas To upload electronic version of your program

The maximum point for this assignment is 10 points. The following points will be deducted if :

- Incorrect file format such as uploading .cbp instead of .cpp , missing electronic copy , using .h and .cpp files , Using list / forward list / stack / queue libraries, Using arrays or global arrays , vectors arrays or global vector arrays , using global variables , not using classes , using recursion, using any concept that was discussed in the class. **(- 10 points)**
- Compilation Errors **(- 8.5 points)**

Other **(at least 1 points each)** if any of the following takes a place :

- Logical Errors
- Lists are not displayed after each function A - I
- Incorrect program file name.
- Not using at least 14 functions.
- Incorrect Style **such as but not limited to** missing comments or program documentations, missing section number, missing function prototypes, missing header / signature line, not replacing my name with your name, output is hard to follow, missing header comments, missing footer comments.